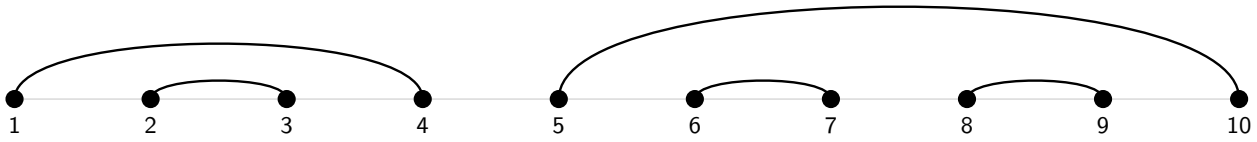
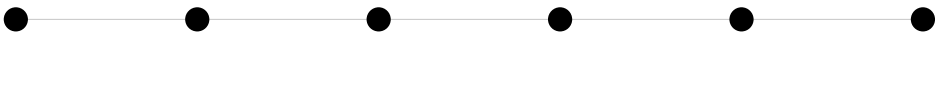
Week 12 / Catalan bijections / Grades 2–3

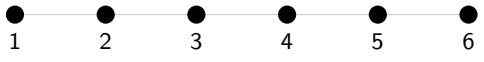
Use each dot once. Keep strings above the row without crossings. Keep dots in place. Pairings are different when a dot has a different partner.

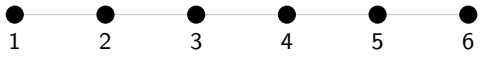
Example with ten dots:

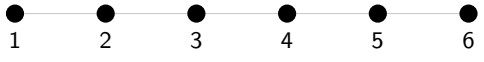


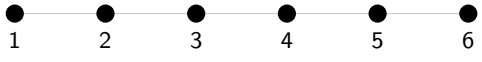
Problem 1: Find every way to pair the six dots. Draw a complete collection with no repeats.









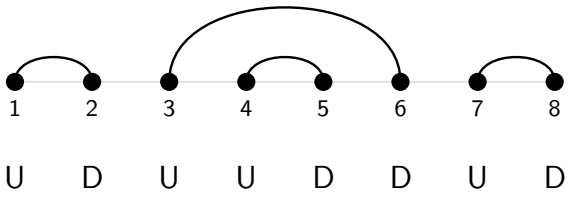


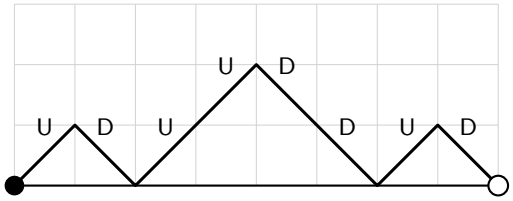




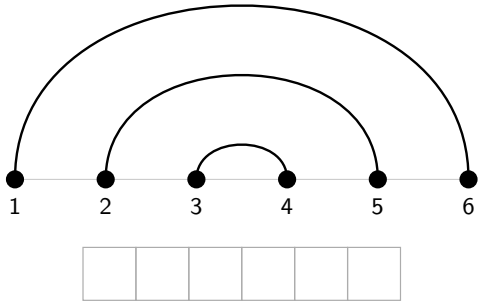
Read the dots from left to right. Write U at the first end of each pair and D at the other end. U is one step up and right; D is one step down and right.

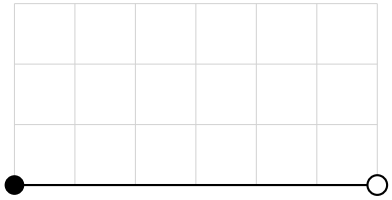
Example:

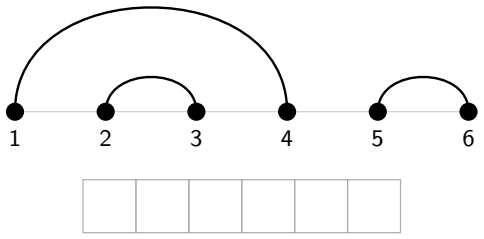


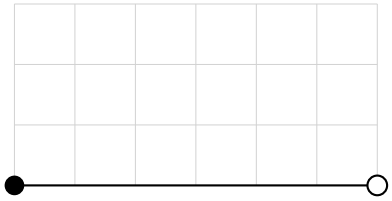


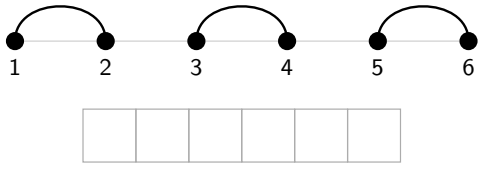
Problem 2: Draw the path for each pairing, starting at the marked dot. Can a pairing ever make a path that goes below the dark line?

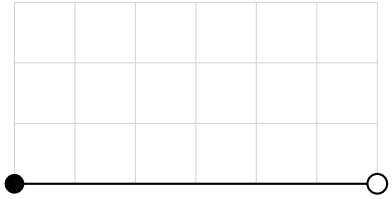




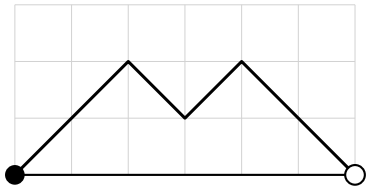




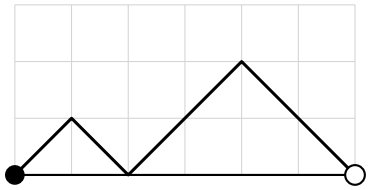




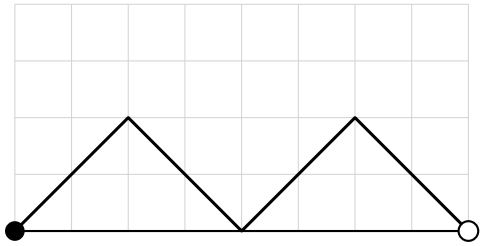
Problem 3: Draw a pairing for each path. Could a path come from two different pairings without crossings? Explain.



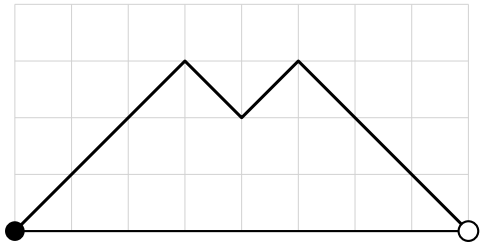






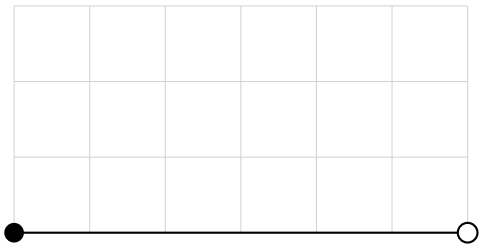


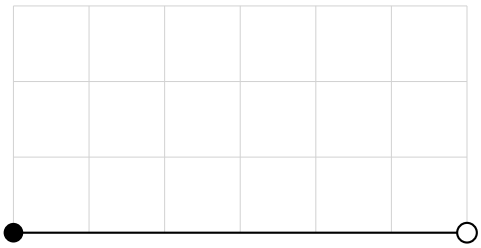


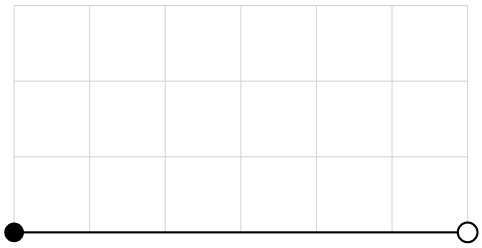


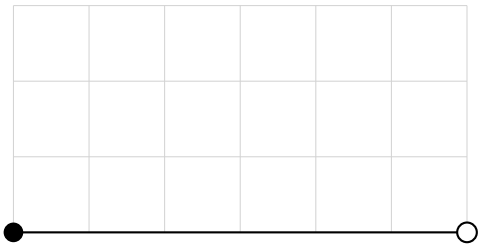


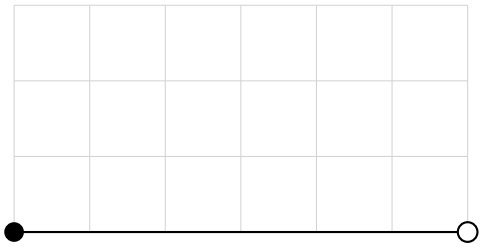
Problem 4: Find every path with three up steps and three down steps that starts and ends on the dark line and never goes below it. Match your paths to your six-dot pairings.

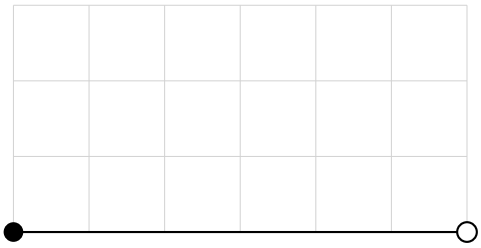












Problem 5: Which strings of letters can be codes for pairings of eight dots? Draw the pairing whenever it is possible, and explain what goes wrong whenever it is impossible.

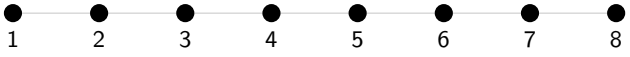
UUUDDUDD



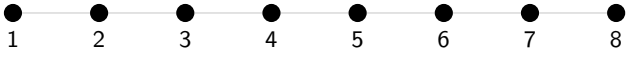
UDDUUDUD



UUDUDUDD



UUUUDDUD



Problem 6: Draw every pairing of eight dots. How can you be sure that none are missing?

